Q1 - 25 July - Shift 1

The slope of the tangent to a curve C : y =y(x) at any point [x, y) on it is $\frac{2e^{2x}-6e^{-x}+9}{2+9e^{-2x}}$. If C passes through the points $\left(0,\frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha,\frac{1}{2}e^{2\alpha}\right)$ then e^{α} is equal to :

Space for your notes:

(A) $\frac{3+\sqrt{2}}{3-\sqrt{2}}$

(B) $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

(D) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Q2 - 25 July - Shift 1

The general solution of the differential equation $(x-y^2)dx+y(5x+y^2)dy=0$ is :

Space for your notes:

(A) $(y^2+x)^4=C|(y^2+2x)^3|$

(B) $(y^2+2x)^4=C|(y^2+x)^3|$

(C) $|(y^2+x)^3|=C(2y^2+x)^4$

(D) $|(y^2+2x)^3|=C(2y^2+x)^4$

Q3 - 25 July - Shift 2

Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the point $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to

(A) $2\log_e 2$ (B) 4

(C) 1 (D) $4\log_e 2$

Space for your notes:

Q4 - 25 July - Shift 2

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1$. If for some $n \in N, y(2) \in [n-1, n)$, then n is equal to ________

Space for your notes:

Q5 - 26 July - Shift 1

If $\frac{dy}{dx} + 2y \tan x = \sin x, 0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is

(A) $\frac{1}{8}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Space for your notes:

Q6 - 26 July - Shift 1

Let a curve $y = y(x)$ pass through the point (3, 3) and the area of the region under this curve, above the x-axis and between the abscissae 3 and x(>3) be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $\left(\alpha, 6\sqrt{10}\right)$ in the first quadrant, then α is equal to ______

Space for your notes:

Q7 - 26 July - Shift 2

Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}, x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

(A) $\frac{\pi}{3} - \frac{1}{4}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Space for your notes:

Q8 - 26 July - Shift 2

Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim_{x \to \infty} y(x)$ is finite. If a and b are respectively the x- and y- intercepts of the tangent to the curve at x=0, then the value of a-4b is equal to _________.

Space for your notes:

Q9 - 27 July - Shift 1

Let $y = y_1(x)$ and $y = y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx} = x + y$, with $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then, the number of points of intersection of $y = y_1(x)$ and $y = y_2(x)$ is

(A) 0
(B) 1
(C) 2
(D) 3

Space for your notes:

Q10 - 27 July - Shift 1

Let $y = y(x)$ be the solution curve of the differential equation

$$\sin\left(2x^2\right)\log_e\left(\tan x^2\right)dy + \left(4xy - 4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)\right)dx = 0,$$

$0 < x < \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}}, 1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to _______.

Space for your notes:

Q11 - 28 July - Shift 1

Let the solution curve of the differential equation $xdy = \left(\sqrt{x^2+y^2}+y\right)dx,\ x>0$, intersect the line $x=1$ at $y=0$ and the line $x=2$ at $y=\alpha$. Then the value of α is :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Space for your notes:

Q12 - 28 July - Shift 1

If $y=y(x),\ x\in\left(0,\frac{\pi}{2}\right)$ be the solution curve of the differential equation

$$\left(\sin^2 2x\right)\frac{dy}{dx}+\left(8\sin^2 2x+2\sin 4x\right)y = 2e^{-4x}\left(2\sin 2x+\cos 2x\right), \quad \text{with} \quad y\left(\frac{\pi}{4}\right)=e^{-\pi},$$

then $y\left(\frac{\pi}{6}\right)$ is equal to :

(A) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}e^{2\pi/3}$

(C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Space for your notes:

Q13 - 28 July - Shift 2

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2 - 1} y = \left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$, $x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7} y(8)$ is equal to

(A) $11 + 6\log_e 3$ (B) 19

(C) $12 - 2\log_e 3$ (D) $19 - 6\log_e 3$

Space for your notes:

Q14 - 28 July - Shift 2

The differential equation of the family of circles passing through the points (0, 2) and (0, – 2) is

(A) $2xy\frac{dy}{dx} + (x^2 - y^2 + 4) = 0$

(B) $2xy\frac{dy}{dx} + (x^2 + y^2 - 4) = 0$

(C) $2xy\frac{dy}{dx} + (y^2 - x^2 + 4) = 0$

(D) $2xy\frac{dy}{dx} - (x^2 - y^2 + 4) = 0$

Space for your notes:

Q15 - 29 July - Shift 1

Let the solution curve $y = y(x)$ of the differential equation $(1 + e^{2x})\left(\frac{dy}{dx} + y\right) = 1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim_{x\to\infty} e^x y(x)$ is equal to :

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

Space for your notes:

Q16 - 29 July - Shift 2

If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passes through the point (2,1) and (k + 1,2), k > 0, then

(A) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$

(B) $\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$

(C) $2\tan^{-1}\left(\frac{1}{k+1}\right) = \log_e\left(k^2+2k+2\right)$

(D) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(\frac{k^2+1}{k^2}\right)$

Space for your notes:

Q17 - 29 July - Shift 2

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \left(\frac{2x^2 + 11x + 13}{x^3 + 6x^2 + 11x + 6}\right) y = \frac{(x+3)}{x+1}, x > -1$, which passes through the point $(0,1)$. Then $y(1)$ is equal to:

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Space for your notes:

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (B)	**Q4** (3)
Q5 (A)	**Q6** (6)	**Q7** (B)	**Q8** (3)
Q9 (A)	**Q10** (1)	**Q11** (B)	**Q12** (A)
Q13 (D)	**Q14** (A)	**Q15** (B)	**Q16** (A)
Q17 (B)			

Q1 (B)

$$\frac{dy}{dx} = \frac{2e^{2x} - 6e^{-x} + 9}{2 + 9e^{-2x}}$$

$$\frac{dy}{dx} = e^{2x} - \frac{6e^{x}}{2e^{2x} + 9}$$

$$y = \frac{e^{2x}}{2} - \tan^{-1}\left(\frac{\sqrt{2}e^{x}}{3}\right) + c$$

If C passes through the point $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$

$$c = -\frac{\pi}{4} - \tan^{-1}\frac{\sqrt{2}}{3}$$

Again C passes through the point $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$

then $e^{\alpha} = \frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

Q2 (A)

$$(x - y^2)dx + y(5x + y^2)dy = 0$$

$$\frac{dy}{dx} = \frac{y^2 - x}{y(5x + y^2)}. \quad \text{Let } y^2 = v$$

$$\frac{2ydy}{dx} = 2\left(\frac{y^2 - x}{5x + y^2}\right)$$

$$\frac{dv}{dx} = 2\left(\frac{v - x}{5x + v}\right) \qquad v = kx$$

$$k + x\frac{dk}{dx} = 2\left(\frac{kx - x}{5x + kx}\right)$$

$$x\frac{dk}{dx} = -\frac{(k^2 + 3k + 2)}{k + 5}$$

$$\int \frac{(5 + k)}{(k + 1)(k + 2)} dk = \int -\frac{dx}{x}$$

$$\int \left(\frac{4}{k + 1} - \frac{3}{k + 2}\right) dk = -\int \frac{dx}{x}$$

$$4 \ln (k + 1) - 3 \ln(k + 2) = -\ln x + \ln c$$

$$\frac{(k + 1)^4}{(k + 2)^3} = -\ln x + \ln c$$

$$c(y^2 + 2x)^3 = (y^2 + x)^4$$

Q3 (B)

(B)

Q4 (3)

(3)

Q5 (A)

$$\frac{dy}{dx} + 2y\tan x = \sin x$$

$$I.F = e^{\int 2\tan x\,dx} = e^{\ln(\sec x)^2} = \sec^2 x$$

$$y\left(\sec^2 x\right) = \int \sin x \sec^2 x\,dx + C$$

$$y.\sec^2 x = \sec x + C$$

$$\text{Put } x = \frac{\pi}{3},\ y = 0$$

$$y = \cos x - 2\cos^2 x$$

$$= \frac{1}{8} - 2\left(\cos x - \frac{1}{4}\right)^2$$

$$\therefore y_{max} = \frac{1}{8}$$

Q6 (6)

$$x^4 = 3yx.y' - 3y^2$$

$$\Rightarrow 3xy\frac{dy}{dx} = 3y^2 + x^4$$

Put $y^2 = t, y\frac{dy}{dx} = \frac{1}{2}\frac{dt}{dx}$

$$\frac{dt}{dx} - \frac{2}{x}t = \frac{2}{3}x^3$$

$$\therefore \frac{t}{x^2} = \frac{x^2}{3} + C$$

$$\Rightarrow \frac{y^2}{x^2} = \frac{x^2}{3} - 2$$

Put $(3, 3)$, $C = -2$

$$\therefore \frac{y^2}{x^2} = \frac{x^2}{3} - 2$$

$$3y^2 = x^4 - 6x^2$$

$$x^4 - 6x^2 = 1080$$

$$\therefore x = 6$$

Q7 (B)

$$\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{x^4+2x}{\sqrt{1-x^2}}$$

$$\text{I.F} = e^{\int \frac{x}{x^2-1}dx}$$

$$\text{I.F} = \sqrt{1-x^2}$$

Solution of D.E.

$$y\cdot\sqrt{1-x^2}=\int\frac{x^4+2x}{\sqrt{1-x^2}}\cdot\sqrt{1-x^2}\,dx$$

$$y\cdot\sqrt{1-x^2}=\int\left(x^4+2x\right)dx$$

$$y\cdot\sqrt{1-x^2}=\frac{x^5}{5}+x^2+C$$

At $x=0$, $y=0$, get $C=0$

$$y=\frac{x^5}{5\sqrt{1-x^2}}+\frac{x^2}{\sqrt{1-x^2}}$$

Now,

$$\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx=\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}}\frac{x^5}{5\sqrt{1-x^2}}dx+\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}}\frac{x^2}{\sqrt{1-x^2}}dx$$

$$\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx=0+2\int_{0}^{\frac{\sqrt{3}}{2}}\frac{x^2}{\sqrt{1-x^2}}dx$$

$$\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx=\frac{\pi}{3}-\frac{\sqrt{3}}{4}$$

Q8 (3)

$$\frac{dy}{dx}-y=2-e^{-x}$$

$$\text{I.F.} = e^{-\int dx}=e^{-x}$$

$\therefore$ solution of D.E

$$y\cdot e^{-x}=\int\left(2e^{-x}-e^{-2x}\right)dx$$

$$\Rightarrow y=-2+\frac{e^{-x}}{2}+C.e^{x}$$

$\because \lim\limits_{x\to\infty} y$ is finite

$$\therefore \lim_{x\to\infty}\left(-2+\frac{e^{-x}}{2}+C.e^{x}\right)\to \text{finite}$$

This is possible only when $C = 0$

$$\therefore y=y(x)=-2+\frac{e^{-x}}{2}$$

$$\frac{dy}{dx}=-\frac{1}{2}e^{-x}$$

$$\left.\frac{dy}{dx}\right|_{x=0}=-\frac{1}{2}=m,\ \ y(0)=-2+\frac{1}{2}=\frac{-3}{2}$$

$\therefore$ equation of tangent

$$y+\frac{3}{2}=-\frac{1}{2}(x-0)$$

$\Rightarrow x + 2y = -3$

$a = -3,\ b = \dfrac{-3}{2}$

$a - 4b = -3 + 6 = 3$

Q9 (A)

$\dfrac{dy}{dx} = x + y \Rightarrow \dfrac{dy}{dx} - y = x$

$\text{I.f} = e^{-x}$

$\therefore$ solution is $ye^{-x} = \int xe^{-x} dx$

$\Rightarrow ye^{-x} = -xe^{-x} - e^{-x} + c$

$\Rightarrow y = -x - 1 + ce^{x}$

$y_1(0) = 0 \Rightarrow c = 1$

$\therefore y_1 = -x - 1 + e^{x} \qquad \ldots.(1)$

$y_2(0) = 1 \Rightarrow c = 2$

$\therefore y_2 = -x - 1 + 2e^{x} \qquad \ldots..(2)$

Now $y_2 - y_1 = e^{x} > 0 \therefore y_2 \neq y_1$

$\therefore$ Number of points of intersection of y_1 & y_2 is zero.

Q10 (1)

$$\sin(2x^2)\ln(\tan x^2)\,dy + \left(4xy - 4\sqrt{2}x\sin\left(x^2 - \frac{\pi}{4}\right)\right)dx = 0$$

$$\ln(\tan x^2)\,dy + \frac{4xy\,dx}{\sin(2x^2)} - \frac{4\sqrt{2}\,x\sin\left(x^2 - \frac{\pi}{4}\right)}{\sin(2x^2)}dx = 0$$

$$d\left(y.\ln(\tan x^2)\right) - 4\sqrt{2}x\frac{(\sin x^2 - \cos x^2)}{\sqrt{2} - 2\sin x^2\cos x^2}dx = 0$$

$$d\left(y.\ln(\tan x^2)\right) - \frac{4x(\sin x^2 - \cos x^2)}{\ldots}dx = 0$$

$$\ldots \frac{[illegible]}{\left(\sin x^2+\cos^2\right)-1}\,dx=0$$

$$\Rightarrow \int d\left(y\ln\left(\tan x^2\right)\right)+2\int\frac{dt}{t^2-1}=\int 0$$

$$\Rightarrow y\ln\left(\tan x^2\right)+2.\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|=c$$

$$y\ln\left(\tan x^2\right)+\ln\left(\frac{\sin x^2+\cos x^2-1}{\sin x^2+\cos x^2+1}\right)=c$$

Put y = 1 and $x=\sqrt{\frac{\pi}{6}}$

$$1\ln\left(\frac{1}{\sqrt{3}}\right)+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}=c$$

Now

$$x=\sqrt{\frac{\pi}{3}}\Rightarrow y\left(\ln\sqrt{3}\right)+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}=\ln\left(\frac{1}{\sqrt{3}}\right)+\ln\left(\frac{\sqrt{3}-1}{\sqrt{3}+3}\right)$$

$$y\left(\ln\sqrt{3}\right)=\ln\left(\frac{1}{\sqrt{3}}\right)$$

$\Rightarrow$ y = -1

$$|y|=1$$

Q11 (B)

$$\mathrm{xdy}=\left(\sqrt{\mathrm{x}^2+\mathrm{y}^2}+\mathrm{y}\right)\mathrm{dx}$$

$$\mathrm{xdy}-\mathrm{ydx}=\sqrt{\mathrm{x}^2+\mathrm{y}^2}\,\mathrm{dx}$$

$$\frac{x\,dy - y\,dx}{x^2} = \sqrt{1 + \frac{y^2}{x^2}} \cdot \frac{dx}{x}$$

$$\frac{d(y/x)}{\sqrt{1 + \left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$$

$$\ln\left(\frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 + 1}\right) = \ln x + R$$

$$\frac{y + \sqrt{y^2 + x^2}}{x} = cx$$

$$y + \sqrt{y^2 + x^2} = cx^2$$

$$x = 1, y = 0 \Rightarrow 0 + 1 = C \Rightarrow C = 1$$

Curve is $y + \sqrt{x^2 + y^2} = x^2$

$$x = 2, y = \alpha$$

$$2 + \sqrt{4 + \alpha^2} = 4$$

$$4 + \alpha^2 = 16 + \alpha^2 = 8\alpha$$

$$\alpha = \frac{3}{2}$$

Q12 (A)

Given differential equation can be re-written as

$$\frac{dy}{dx}+\left(8+4\cot 2x\right)y=\frac{2e^{-4x}}{\sin^2 2x}\left(2\sin x+\cos 2x\right)$$

which is a linear diff. equation.

$$\text{I.f.} = e^{\int(8+4\cot 2x)dx} = e^{8x+2Cu(\sin 2x)}$$

$$= e^{8x}\cdot\sin^2 2x$$

$\therefore$ solution is

$$y\left(e^{8x}\cdot\sin^2 2x\right)=\int 2e^{4x}\left(2\sin 2x+\cos 2x\right)dx+C$$

$$= e^{4x}\cdot\sin 2x + C$$

Given $y\left(\frac{\pi}{4}\right)=e^{-\pi}\Rightarrow C=0$

$$\therefore\ y=\frac{e^{-4x}}{\sin 2x}$$

$$\therefore\ y\left(\frac{\pi}{6}\right)=\frac{e^{-4\cdot\frac{\pi}{6}}}{\sin\left(2\cdot\frac{\pi}{6}\right)}=\frac{2}{\sqrt{3}}e^{-\frac{2\pi}{3}}$$

Q13 (D)

$$\frac{dy}{dx}+\frac{1}{x^2-1}y=\left(\frac{x-1}{x+1}\right)^{\frac{1}{2}},$$

$$\frac{dy}{dx}+Py=Q$$

$$\text{I.F.}=e^{\int Pdx}=\left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$$

$$y\left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}=\int\left(\frac{x-1}{x+1}\right)^{1}dx$$

$$=x-2\log_e|x+1|+C$$

Curve passes through $\left(2,\frac{1}{\sqrt{3}}\right)$

$$\Rightarrow C=2\log_e 3-\frac{5}{3}$$

at $x=8$,

$$\sqrt{7}y(8)=19-6\log_e 3$$

Q14 (A)

, Equation of circle passing through (0, –2) and (0, 2) is

$$x^2+\left(y^2-4\right)+\lambda x=0,\ (\lambda \in R)$$

Divided by x we get

$$\frac{x^2+\left(y^2-4\right)}{x}+\lambda=0$$

Differentiating with respect to x

$$\frac{x\left[2x+2y\cdot\frac{dy}{dx}\right]-\left[x^2+y^2-4\right]\cdot 1}{x^2}=0$$

$$\Rightarrow 2xy\cdot\frac{dy}{dx}+\left(x^2-y^2+4\right)=0$$

Q15 (B)

$$\frac{dy}{dx}+y=\frac{1}{1+e^{2x}}$$

So integrating factor is $e^{\int 1.dx}=e^{x}$

So solution is $y\cdot e^{x}=\tan^{-1}\left(e^{x}\right)+c$

Now as curve is passing through $\left(0,\frac{\pi}{2}\right)$ so

$$\Rightarrow c=\frac{\pi}{4}$$

$$\Rightarrow \lim_{x\to\infty}\left(y\cdot e^{x}\right)=\lim_{x\to\infty}\left(\tan^{-1}\left(e^{x}\right)+\frac{\pi}{4}\right)=\frac{3\pi}{4}$$

Q16 (A)

$$\frac{dy}{dx}=\frac{x+y-2}{x-y}=\frac{(x-1)+(y-1)}{(x-1)-(y-1)}$$

$x-1=X,\ y-1=Y$

$$\frac{dY}{dX}=\frac{X+Y}{X-Y}$$

$$Y=VX \qquad \frac{dY}{dX}=V+X\frac{dV}{dX}$$

$$V+X\frac{dV}{dX}=\frac{1+V}{1-V} \qquad X\frac{dV}{dX}=\frac{V^2+1}{1-V}$$

$$\int\frac{1-V}{1+V^2}dV=\int\frac{dX}{X}$$

$$\int\frac{dV}{1+V^2}-\frac{1}{2}\int\frac{2VdV}{1+V^2}=\int\frac{dX}{X}$$

$$\tan^{-1}V-\frac{1}{2}\ln\left(1+V^2\right)=\ln X+c$$

$$\tan^{-1}\left(\frac{Y}{X}\right)-\frac{1}{2}\ln\left(1+\frac{Y^2}{X^2}\right)=\ln(X)+c$$

$$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ln\left(1+\frac{(y-1)^2}{(x-1)^2}\right)=\ln(x-1)+c$$

Passes through (2,1)

$$0-\frac{1}{2}\ \ln 1=\ln 1+c\ \therefore\ c=0$$

Passes through (k + 1, 2)

$$\therefore\ \tan^{-1}\left(\frac{1}{k}\right)-\frac{1}{2}\text{In}\left(1+\frac{1}{k^2}\right)=\ln k$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\text{In}\left(\frac{1+k^2}{k^2}\right)+2\ln k$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\ln(1+k^2)$$

Q17 (B)

$$\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{x+3}{x+1}$$

$$\int p(x)dx \qquad\qquad \text{I.F.}=e^{\int p(x)dx}$$

$$\int p(x)dx=\int\frac{\left(2x^2+11x+13\right)dx}{(x+1)(x+2)(x+3)}$$

Using partial fraction

$$\frac{2x^2+11x+13}{(x+1)(x+2)(x+3)}=\frac{A}{x+1}+\frac{B}{x+2}+\frac{C}{x+3}$$

$$A=\frac{4}{2}=2$$

$B = 1$

$C = -1$

$$\because \int p(x)dx=A\ln(x+1)+B\ln(x+2)+c\ln(x+3)$$

$$=\ln\left(\frac{(x+1)^2(x+2)}{x+3}\right)$$

$$\text{I.F.}=e^{\int p(x)dx}=\frac{(x+1)^2(x+2)}{(x+3)}$$

Solution $y(\text{IF})=\int Q.(\text{IF})dx$

$$y\left(\frac{(x+1)^2(x+2)}{x+3}\right)=\int\left(\frac{\cancel{x+3}}{\cancel{x+1}}\right)\frac{(x+1)^{\cancel{2}}(x+2)}{(\cancel{x+3})}dx$$

$$y\left(\frac{(x+1)^2(x+2)}{x+3}\right)=\frac{x^3}{3}+\frac{3x^2}{2}+2x+c$$

Passes through (0, 1) $C=\frac{2}{3}$

Now put x = 1

$$\Rightarrow y(1)=\frac{3}{2}$$